SAGAR PUBLIC SCHOOL

Integrated Programme • Topic: Gravitation

PRACTICE SHEET – 1

Newton's Law of Gravitation, Gravitational Field and Superposition

Name: _____	Class / Section: _____
Date: _____	Roll No.: _____
Time Allowed: 60 Mins	**Total Questions: 20**

Instructions: Attempt all questions. Section A has only one correct option; Section B may have more than one correct option; Section C needs full working. Time Allowed: 60 Mins.

Section A – Single Correct Option (Q.1 to Q.11)

Q.1 At what altitude will the acceleration due to gravity be 25% of that at the earth's surface (given radius of earth is R)?

(A) $R/4$ (B) R (C) $3R/8$ (D) $R/2$

[Motion, Ex-I Q.1]

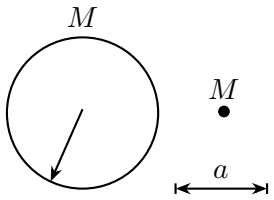
Q.2 Two masses m_1 & m_2 are initially at rest and are separated by a very large distance. If the masses approach each other subsequently, due to gravitational attraction between them, their relative velocity of approach at a separation distance of d is

(A) $\dfrac{2Gd}{(m_1+m_2)}$ (B) $\dfrac{(m_1+m_2)G}{2d}$

(C) $\left[(m_1+m_2)\dfrac{2G}{d}\right]^{1/2}$ (D) $(m_1+m_2)^{1/2}\,2Gd$

[Motion, Ex-I Q.2]

Q.3 A particle of mass M is at a distance a from surface of a thin spherical shell of equal mass and having radius a.



(A) Gravitational field and potential both are zero at centre of the shell
(B) Gravitational field is zero not only inside the shell but at a point outside the shell also
(C) Inside the shell, gravitational field alone is zero
(D) Neither gravitational field nor gravitational potential is zero inside the shell

[Motion, Ex-I Q.7]

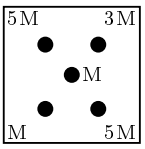
Q.4 A hollow spherical shell is compressed to half its radius. The gravitational potential at the centre

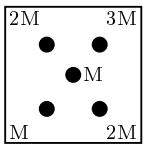
(A) increases (B) decreases

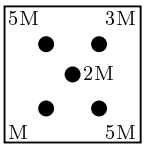
(C) remains same (D) during the compression increases then returns at the previous value

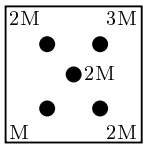
[Motion, Ex-I Q.11]

Q.5 A mass is at the center of a square, with four masses at the corners as shown.

(A) 

(B) 

(C) 

(D) 

Rank the choices according to the magnitude of the gravitational force on the center mass.

(A) $F_A = F_B < F_C = F_D$ (B) $F_A > F_B < F_D < F_C$

(C) $F_A = F_B > F_C = F_D$ (D) none

[Motion, Ex-I Q.16]

Q.6 Choose the **INCORRECT** option.

(A) Inertial mass is a measure of difficulty of accelerating a body by an external force whereas the gravitational mass is relevant in determining the gravitational force on it by an external mass.

(B) That the gravitational mass and inertia mass are equal is an experimental result.

(C) That the acceleration due to gravity on earth is the same for all bodies is due to the equality of gravitational mass and inertia mass.

(D) Gravitational mass of a particle like proton can depend on the present of neighbouring heavy objects but the inertia mass cannot.

[Nucleus, O-1 Q.1]

Q.7 If the distance between the centres of Earth and Moon is D and mass of Earth is 81 times that of Moon. At what distance from the centre of Earth gravitational field will be zero?

(A) $\frac{D}{2}$ (B) $\frac{2D}{3}$ (C) $\frac{4D}{5}$ (D) $\frac{9D}{10}$

[Nucleus, O-1 Q.2]

Q.8 A solid sphere of mass M and radius a is surrounded by a uniform concentric spherical shell of thickness $2a$ and mass $2M$. The gravitational field at distance $3a$ from the centre will be:

(1) $\frac{GM}{3a^2}$ (2) $\frac{2GM}{9a^2}$ (3) $\frac{GM}{9a^2}$ (4) $\frac{2GM}{3a^2}$

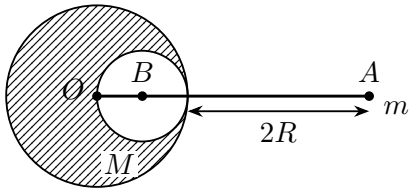
[Nucleus, JM Q.16 [JEE Main-2019]]

Q.9 Two stars of masses m and $2m$ at a distance d rotate about their common centre of mass in free space. The period of revolution is:

(1) $\frac{1}{2\pi}\sqrt{\frac{d^3}{3Gm}}$ (2) $2\pi\sqrt{\frac{d^3}{3Gm}}$ (3) $\frac{1}{2\pi}\sqrt{\frac{3Gm}{d^3}}$ (4) $2\pi\sqrt{\frac{3Gm}{d^3}}$

[Nucleus, JM Q.31 [JEE Main-2021]]

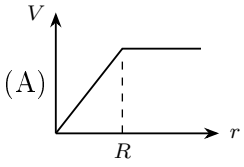
Q.10 A solid sphere of radius R gravitationally attracts a particle placed at $3R$ from its centre with a force F_1. Now a spherical cavity of radius $\left(\frac{R}{2}\right)$ is made in the sphere (as shown in figure) and the force becomes F_2. The value of $F_1 : F_2$ is:

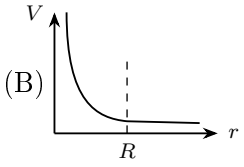


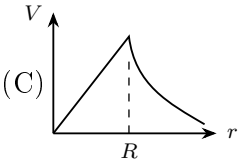
(1) 25 : 36 (2) 36 : 25 (3) 50 : 41 (4) 41 : 50

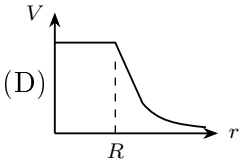
[Nucleus, JM Q.36 [JEE Main-2021]]

Q.11 A spherically symmetric gravitational system of particles has a mass density $\rho = \begin{cases} \rho_0 & \text{for } r \le R \\ 0 & \text{for } r > R \end{cases}$ where ρ_0 is a constant. A test mass can undergo circular motion under the influence of the gravitational field of particles. Its speed V as a function of distance r $(0 < r < \infty)$ from the centre of the system is represented by

(A) 

(B) 

(C) 

(D) 

[Nucleus, JA Q.1 [IIT-JEE 2008]]

Section B – One or More Than One Option Correct (Q.12 to Q.13)

Q.12 Assuming the earth to be a sphere of uniform density the acceleration due to gravity

(A) at a point outside the earth is inversely proportional to the square of its distance from the center
(B) at a point outside the earth is inversely proportional to its distance from the centre
(C) at a point inside is zero
(D) at a point inside is proportional to its distance from the centre

[Motion, Ex-II Q.1]

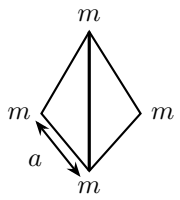
Q.13 Inside a hollow spherical shell

(A) everywhere gravitational potential is zero
(B) everywhere gravitational field is zero
(C) everywhere gravitational potential is same
(D) everywhere gravitational field is same

[Motion, Ex-II Q.3]

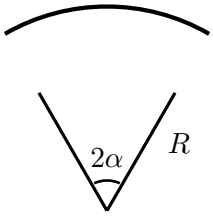
Section C – Subjective / Numerical Problems (Q.14 to Q.20)

Q.14 Four masses (each of m) are placed at the vertices of a regular pyramid (triangular base) of side 'a'. Find the work done by the system while taking them apart so that they form the pyramid of side '$2a$'.



[Motion, Ex-III Q.1]

Q.15 Find the gravitational field strength and potential at the centre of arc of linear mass density λ subtending an angle 2α at the centre.



[Motion, Ex-III Q.6]

Q.16 A point P lies on the axis of a fixed ring of mass M and radius a, at a distance a from its centre C. A small particle starts from P and reaches C under gravitational attraction only. Its speed at C will be _____.

[Motion, Ex-III Q.7]

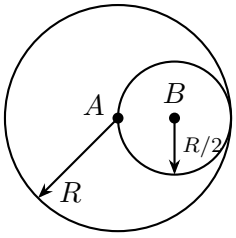
Q.17 Calculate the distance from the surface of the earth at which above and below the surface acceleration due to gravity is the same.

[Motion, Ex-III Q.8]

Q.18 Find the potential energy of a system of eight particles placed at the vertices of a cube of side L. Neglect the self energy of the particles.

[Motion, Ex-III Q.12]

Q.19 Inside an isolated fixed sphere of radius R and uniform density ρ, there is a spherical cavity of radius $R/2$ such that the surface of the cavity passes through the centre of the sphere as in figure. A particle of mass m is released from rest at centre B of the cavity. Calculate velocity with which particle strikes the centre A of the sphere.



[Motion, Ex-IV Q.4]

Q.20 Two uniform spherical stars made of same material have radii R and $2R$. Mass of the smaller planet is m. They start moving from rest towards each other from a large distance under mutual force of gravity. The collision between the stars is inelastic with coefficient of restitution 1/2.
(a) Find the kinetic energy of the system just after the collision.
(b) Find the maximum separation between their centres after their first collision.

[Nucleus, S-1 Q.3]

ANSWER KEY – Practice Sheet 1

Answers are taken from the source answer keys (Motion: Answer Ex-I to Ex-IV; Nucleus: Answer Key page). JM answers are option numbers.

Q.	Source	Answer
1	Motion Ex-I Q.1	B
2	Motion Ex-I Q.2	C
3	Motion Ex-I Q.7	D
4	Motion Ex-I Q.11	B
5	Motion Ex-I Q.16	A
6	Nucleus O-1 Q.1	D
7	Nucleus O-1 Q.2	D
8	Nucleus JM Q.16	(1)
9	Nucleus JM Q.31	(2)
10	Nucleus JM Q.36	(3)
11	Nucleus JA Q.1	C
12	Motion Ex-II Q.1	A, D
13	Motion Ex-II Q.3	B, C, D
14	Motion Ex-III Q.1	$-\dfrac{3Gm^2}{a}$
15	Motion Ex-III Q.6	Field $= \dfrac{2G\lambda}{R}\sin\alpha$; Potential $= -G\lambda\, 2\alpha$
16	Motion Ex-III Q.7	$\sqrt{\dfrac{2GM}{a}\left(1-\dfrac{1}{\sqrt{2}}\right)}$
17	Motion Ex-III Q.8	$h = \dfrac{\sqrt{5}-1}{2}R$
18	Motion Ex-III Q.12	$-\dfrac{4GM^2}{L}\left[3+\dfrac{3}{\sqrt{2}}+\dfrac{1}{\sqrt{3}}\right]$
19	Motion Ex-IV Q.4	$\sqrt{\dfrac{2}{3}\pi G\rho R^2}$
20	Nucleus S-1 Q.3	(a) $\dfrac{2Gm^2}{3R}$; (b) $4R$

SELF ASSESSMENT – Practice Sheet 1

Fill this in after checking your answers with the Answer Key.

Time Allowed: 60 Mins	**Time Taken:** _____ Mins
Start time: _____	End time: _____
Finished within the allowed time? ☐ Yes ☐ No (extra / saved time: _____ Mins)	

Total	Attempted	Correct	Incorrect	Not attempted
20				

Score (Correct / 20): _____ Percentage: _____ %

Question-wise tracker write ✓ (correct), ✗ (incorrect) or – (not attempted)

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20

Reason for the incorrect / unattempted questions (write question numbers)

Concept not clear: _____	Formula forgotten: _____
Calculation / silly mistake: _____	Ran out of time: _____

Personal feedback guide (tick the band that matches your score)

	Correct	What to do next
☐	18–20	Excellent. Concept and speed are both strong. Move to harder JEE-level problems and keep revising formulae once a week.
☐	14–17	Good. Re-do every incorrect question without looking at the solution, then note the formula or idea you missed.
☐	9–13	Fair. Revise the theory of fields, potential and orbits (shell, ring, sphere results), then re-attempt this sheet after 3 days.
☐	0–8	Needs attention. Re-read the chapter notes and solved examples first, then attempt this sheet again in two parts of 30 mins.

My personal feedback / action plan:

Teacher's remarks:

Student's signature: _____ Teacher's signature: _____